Impulse changes momentum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect force over time to momentum change.

SUCCESS CRITERIA - I can...

1. Calculate impulse from force and duration.
2. Match impulse direction to momentum change.
3. Use impulse to find a final velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “impulse” and “average” to explain your answer. Sentence starters are optional.

impulse: net force acting over time; change in momentum

average: one value representing an interval or a set of values

momentum: mass times velocity, including direction

01 Warm-up: bring back an earlier idea

A planet has surface field **75 N/kg**. At center distance **5** R, find field and the force on a **7 kg** object.

Sentence starter: My design includes … . It does what is needed by … .

Marzano check - circle: ① ② ③

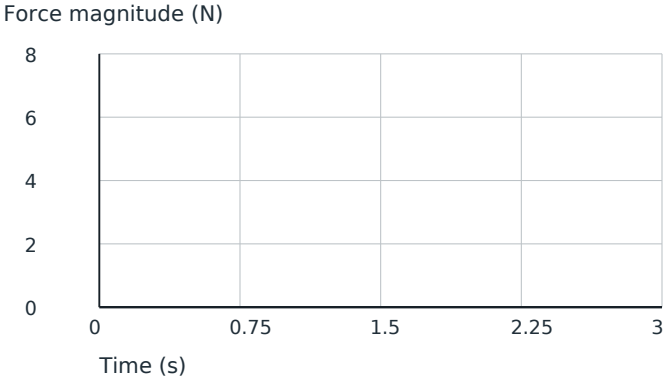
02 Worked example: explain the prediction

Reference: $J = F_{avg}\,\Delta t = \Delta p = p_f - p_i$.
J is impulse; F_{avg} is average net force. Signed force-time area gives impulse; i = initial, f = final.

Worked reasoning: A constant **6 N** force acts right for **2 s**. Impulse is **12 N·s** right, so momentum increases by **12 kg·m/s**.

Your turn: 1. **Draw the force-time rectangle** and **shade its area**.
2. **Explain why the two impulse** units are equivalent.

Sentence starter: I think … . I can point to …, which shows …, so … .

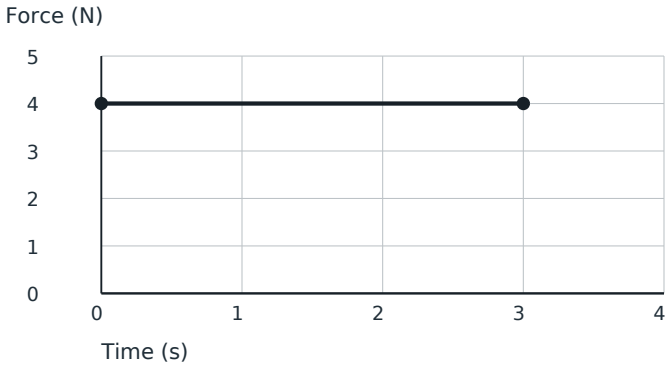


Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Read the rectangular force-time graph**.
2. **Find the area** and **state what it means**.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

04 Apply the physics

A **2 kg** cart starts at rest and receives **6 N·s** rightward impulse. Final momentum is **6 kg·m/s**. **Find final velocity** by dividing by mass.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

Before	**After**

Label mass and velocity.

Marzano check - circle: ① ② ③

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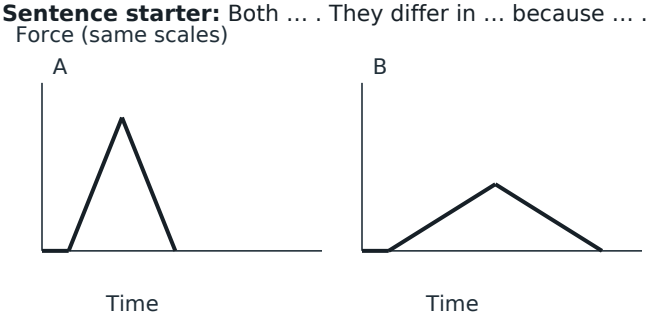
Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Two force pulses have equal signed area but different peak heights. What must be equal? What need not be equal?

Sentence starter: Both … . They differ in … because … .



Same scales: compare impulse and peak force.

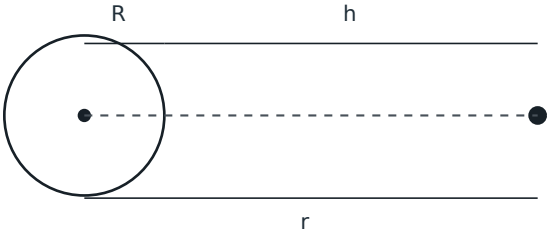
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A spherical planet has surface field **144 N/kg**. **Find the field** at center distance **6** R, where R is the planet radius. **Label the masses**, field or force relevant to the question.

Word help: radius = distance from the center to the edge of a circle or sphere

Sentence starter: My design includes … . It does what is needed by … .



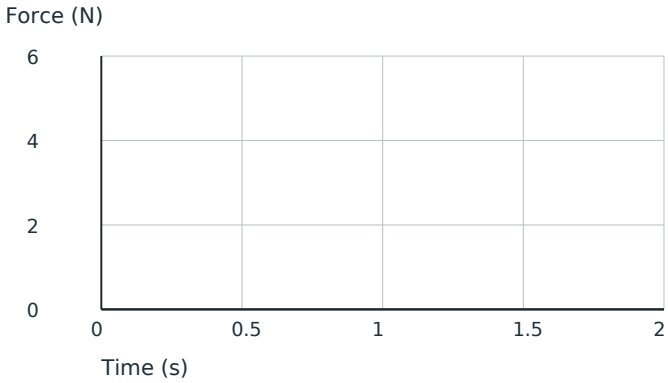
Marzano check - circle: ① ② ③

07 Exit check

A **5 N** force acts right for **2 s**. **Find impulse**. On a **2 kg** cart initially at rest, find final velocity.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I found … [unit]. This value means … in this situation.

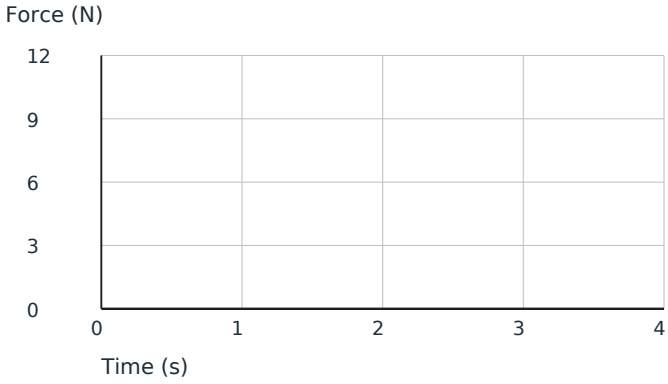


Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

1. Create two constant force-and-time pairs giving an impulse of **12 N·s**.
2. **Compare their force-time rectangles**.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: ① ② ③